

E14 Calculating residence time distribution quantities

This is Fogler Examples 16-1 and 16-2.

A sample of the tracer hythane was injected as a pulse into a reactor and the exit concentration was measured as a function of time. The data is given below.

1. Construct a curve showing $C(t)$.
2. Construct the $E(t)$ curve.
3. Construct the $F(t)$ curve.
4. Determine the mean residence time.
5. Calculate the fraction of fluid that spends between 3 and 6 minutes in the reactor.

t (min)	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
C (g/m ³)	0	0.6	1.4	5	8	10	8	6	4	3	2.2	1.6	0.6	0

Decision Guide: Calculating Residence Time Distribution Quantities

Goal

Given tracer-concentration data $C(t)$, determine:

1. The $C(t)$ curve.
2. The $E(t)$ curve.
3. The $F(t)$ curve.
4. The mean residence time.
5. The fraction of fluid that spends between 3 and 6 minutes in the reactor.

Step 1: Identify the Type of RTD Experiment

The problem states that: A pulse of tracer was injected and the outlet concentration was measured as a function of time. This means that:

$C(t)$ is the starting information.

Everything else is calculated from the concentration data.

Step 2: Construct the $C(t)$ Curve

Plot: Tracer Concentration vs Time using the tabulated data.

This plot is the raw experimental result.

Before doing any calculations:

- Check that the curve starts near zero.
- Check that the curve returns to zero.
- Verify that the peak appears reasonable.

Step 3: Calculate the Area Under the $C(t)$ Curve

The first numerical calculation is:

$$\int_0^{\infty} C(t) dt$$

This area is required to normalize the RTD.

Because the data are discrete, use numerical integration (typically the trapezoidal rule).

Everything that follows depends on this area.

Step 4: Calculate $E(t)$

Use:

$$E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$$

This converts the concentration curve into a probability-density function.

Interpretation: $E(t)dt$ is the fraction of fluid that exits between t and $t+dt$

Step 5: Construct the E(t) Curve

Plot: E(t) vs Time after calculating the normalized values.

Before moving on, verify:

$$\int_0^{\infty} E(t) dt = 1$$

or very close to 1 (small deviations arise from numerical approximation).

Step 6: Calculate the F(t) Curve

The cumulative RTD function is:

$$F(t) = \int_0^t E(t) dt$$

Interpretation:

F(t) is the fraction of fluid that has exited by time t . Calculate cumulative areas moving from left to right across the data table.

Step 7: Construct the F(t) Curve

Plot: F(t) vs Time. The curve should:

- Start at 0.
- Increase continuously.
- Approach 1 at long times.

Step 8: Calculate the Mean Residence Time

Use:

$$\bar{t} = \int_0^{\infty} t \cdot E(t) dt$$

For tabulated data:

Multiply $t \times E(t)$. Then integrate numerically. The mean residence time represents the average amount of time fluid spends in the reactor.

Step 9: Calculate the Fraction Between 3 and 6 Minutes

Use:

$$\int_3^6 E(t) dt$$

Interpretation: Fraction of fluid that spends between 3 and 6 minutes in the reactor.

Alternatively: $F(6) - F(3)$ which is often easier once the $F(t)$ curve has been calculated.

Step 10: Check for Consistency

Verify:

- ✓ $E(t)$ integrates to 1
- ✓ $F(t)$ approaches 1
- ✓ Mean residence time falls within the range of measured times
- ✓ Fraction between 3 and 6 minutes is between 0 and 1

If any of these fail, re-check the numerical integration.

Final Answer Checklist

- ✓ Constructed $C(t)$ curve.
- ✓ Calculated the area under $C(t)$.
- ✓ Constructed $E(t)$ curve.
- ✓ Constructed $F(t)$ curve.
- ✓ Calculated the mean residence time.

- ✓ Calculated the fraction of fluid between 3 and 6 minutes.

CBE 3610 Reactor Design Handbook Equations Needed

No specific RTD equations appear in the Handbook. Use the RTD definitions introduced in lecture:

$$E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$$

$$F(t) = \int_0^t E(t') dt'$$

$$\bar{t} = \int_0^{\infty} t E(t) dt$$

$$\text{Fraction between } t_1 \text{ and } t_2 = \int_{t_1}^{t_2} E(t) dt = F(t_2) - F(t_1)$$

Key Insight

Experimental Data

↓

C(t)

↓

Normalize

↓

E(t)

↓

Integrate

↓

F(t)

↓

Calculate Mean Residence Time and Fluid Fractions

Almost every RTD calculation follows this exact sequence. Once you obtain $E(t)$, all of the remaining RTD quantities become straightforward to calculate.